

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Name	Kamalesh A	Section	CSE-B
Registration No.	3122235001063	Date	03-03-2026

Experiment 7

Bagging, Boosting, and Stacked Ensemble Models

1. Aim and Objective

- To understand ensemble learning strategies: **Bagging**, **Boosting**, and **Stacking**.
- To implement a **Bagging classifier** using a Decision Tree as the base estimator.
- To implement **Boosting classifiers** – AdaBoost and Gradient Boosting.
- To build a **Stacked Ensemble model** using multiple heterogeneous base learners.
- To compare ensemble models in terms of accuracy, stability, and generalization using 5-Fold Cross-Validation.
- To analyze the effect of ensemble methods on bias and variance.

2. Dataset Description

Wisconsin Diagnostic Breast Cancer Dataset

- Total samples: 569
- Features: 30 numerical attributes
- Target classes: Malignant ($M = 1$) and Benign ($B = 0$)
- Class distribution: 357 Benign, 212 Malignant

Dataset Link: <https://archive.ics.uci.edu>

3. Preprocessing Steps

- The dataset was loaded from `wdbc.data` with 569 instances and 32 attributes (`id`, `diagnosis`, and 30 morphological features).
- Feature column names were manually assigned including `id`, `diagnosis`, and 30 numerical morphological features (mean, standard error, and worst values).
- The target variable `diagnosis` was mapped from categorical strings ($M \rightarrow 1$, $B \rightarrow 0$).
- Non-predictive columns (`id`) were dropped, yielding a feature matrix X of shape $(569, 30)$.
- A correlation heatmap was plotted to visualize inter-feature relationships.
- The dataset was split into training and testing sets using an 80–20 ratio with `random.state=42` and stratified sampling to preserve class proportions.

4. Implementation Details

- **Bagging:** A `BaggingClassifier` was implemented using a `DecisionTreeClassifier` as the base estimator. Hyperparameters such as `n_estimators`, `max_samples`, and `max_features` were tuned using `GridSearchCV` with 5-fold cross-validation.
- **AdaBoost:** `AdaBoostClassifier` was implemented and tuned over `n_estimators` and `learning_rate`. The model improves performance by focusing on misclassified samples iteratively.
- **Gradient Boosting:** `GradientBoostingClassifier` was implemented and tuned using `n_estimators`, `learning_rate`, and `max_depth`. It minimizes residual errors sequentially.
- **Stacking:** A `StackingClassifier` was implemented using Decision Tree and KNN as base models, and Logistic Regression as the meta-learner. Different configurations of Logistic Regression (varying C and solver) were evaluated.
- All models were evaluated using **5-Fold Cross-Validation** with metrics **Accuracy** and **F1 Score**.

5. Visualizations

In this section, various visualizations are used to analyze the dataset and evaluate the performance of ensemble models.

Feature Correlation Heatmap

A correlation heatmap is used to visualize relationships between features in the dataset. Highly correlated features indicate redundancy and strong dependencies.

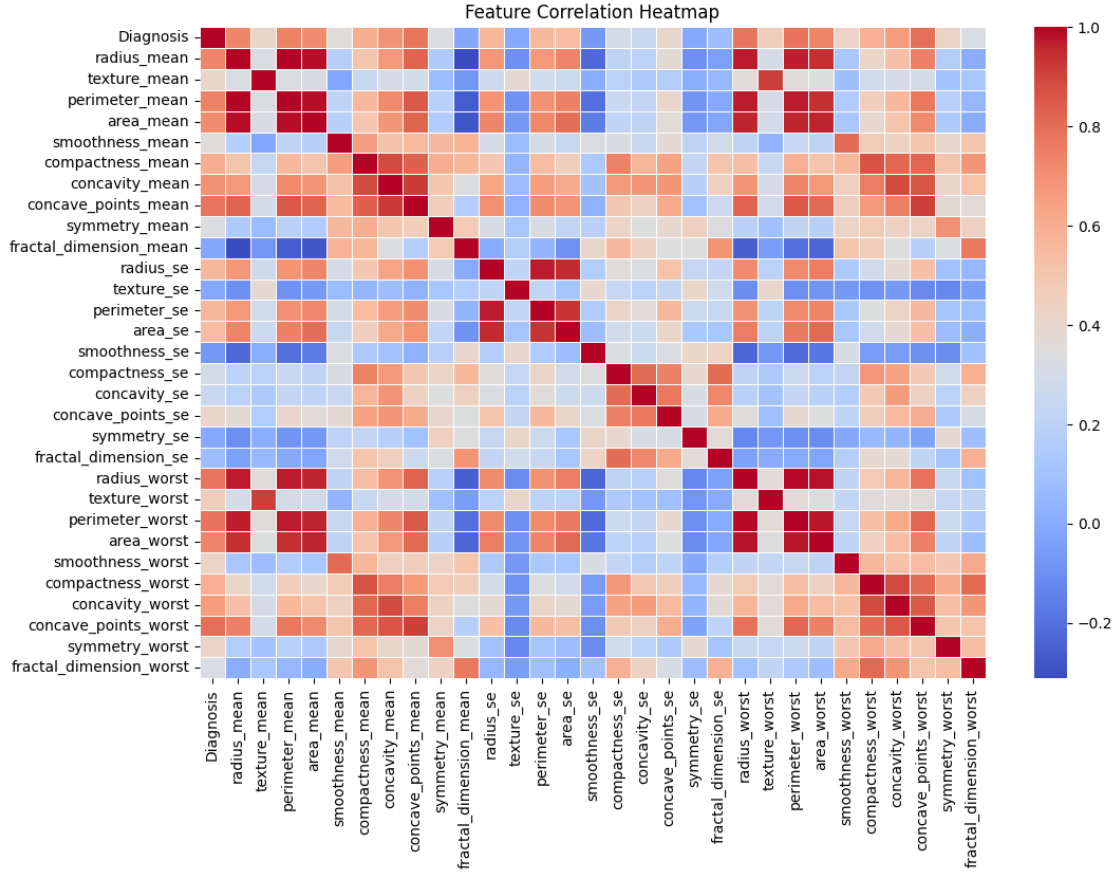


Figure 1: Feature Correlation Heatmap

Confusion Matrices

Confusion matrices are used to evaluate classification performance by comparing actual and predicted values for each model.

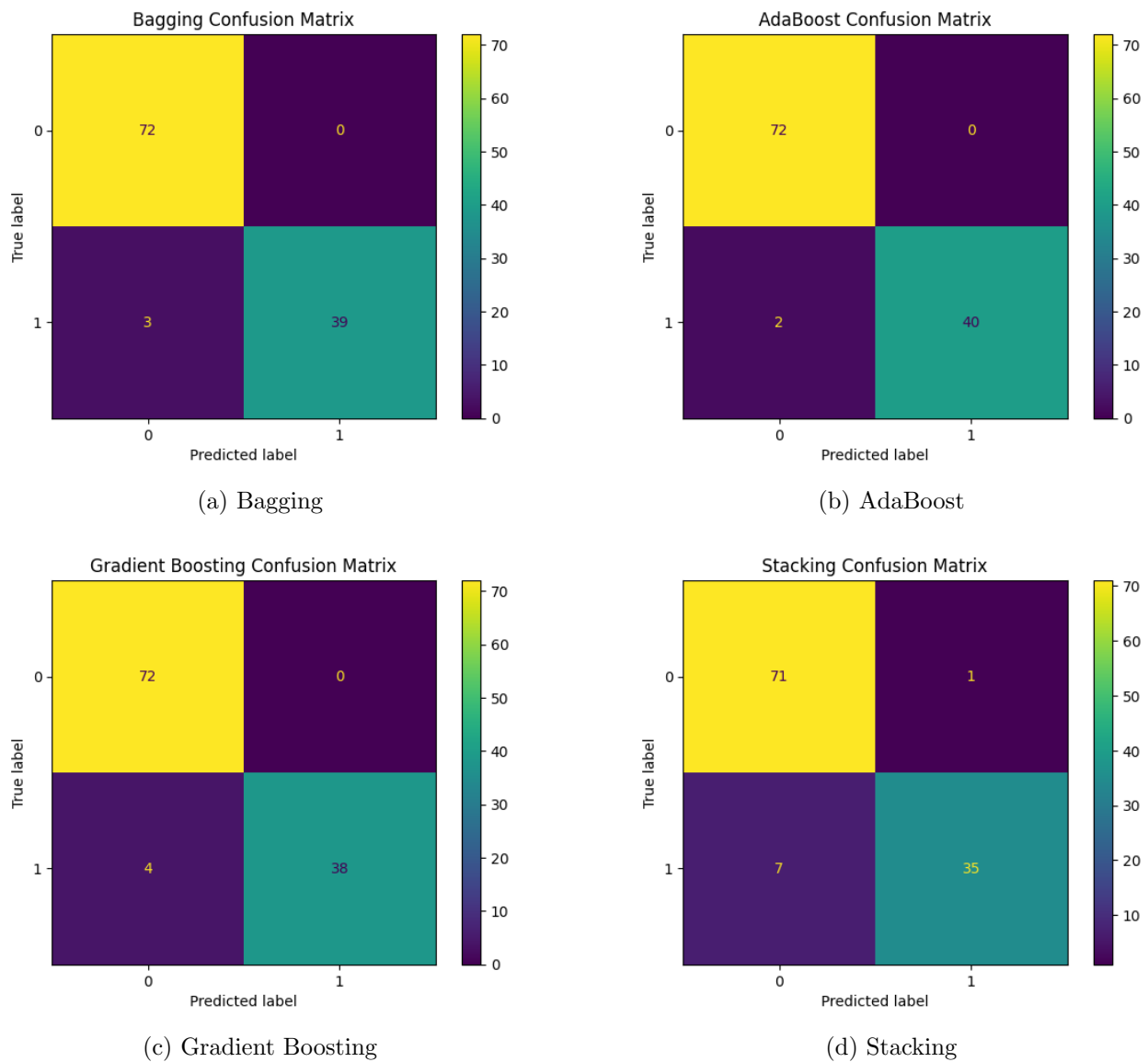


Figure 2: Confusion Matrices for Ensemble Models

ROC Curve and AUC

Receiver Operating Characteristic (ROC) curves are plotted to evaluate the trade-off between True Positive Rate (TPR) and False Positive Rate (FPR). The Area Under Curve (AUC) indicates model performance.

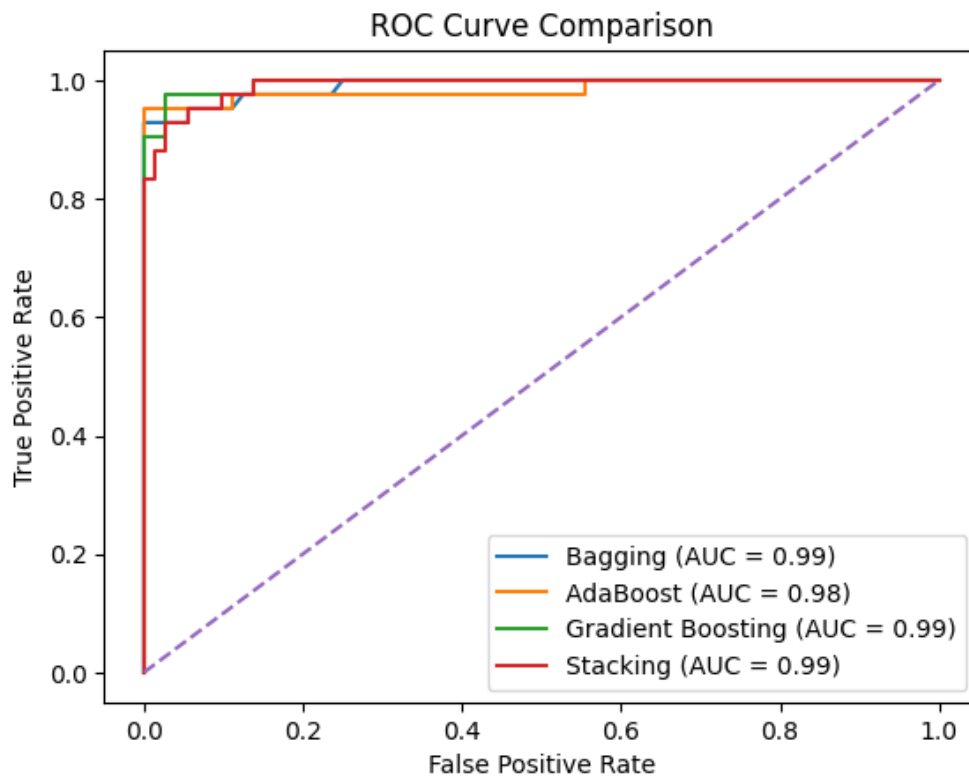


Figure 3: ROC Curve Comparison for Bagging, AdaBoost, Gradient Boosting, and Stacking

6. Hyperparameter Tuning Results

In this section, the performance of Bagging, Boosting, and Stacked Ensemble models is evaluated using 5-Fold Cross-Validation. The models are compared using Average Cross-Validation Accuracy and F1 Score.

Bagging Results

Table 1: Bagging Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	max_samples	max_features	Avg CV Accuracy (%)	Avg CV F1 Score
10	0.6	0.6	94.73	0.9269
50	0.6	0.6	95.60	0.9416
100	0.6	0.6	95.16	0.9344
10	0.8	0.6	95.82	0.9420
50	0.8	0.6	95.60	0.9408
100	1.0	1.0	96.04	0.9487
100	0.8	0.8	96.48	0.9648

Boosting Results

Table 2: Boosting Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	learning_rate	Avg CV Accuracy (%)	Avg CV F1 Score
<i>AdaBoost</i>			
50	0.01	92.31	0.8908
100	0.01	92.53	0.8957
200	0.01	93.19	0.9057
50	0.1	94.95	0.9314
100	0.1	95.16	0.9356
200	0.1	95.16	0.9349
50	1.0	96.26	0.9489
100	1.0	96.04	0.9463
200	1.0	96.48	0.9524
<i>Gradient Boosting</i>			
50	0.05	94.29	0.9223
100	0.05	95.16	0.9345
50	0.05 (depth=5)	93.85	0.9166
100	0.05 (depth=5)	94.07	0.9194
50	0.1 (depth=3)	95.60	0.9404

Stacked Ensemble Results

Table 3: Stacked Ensemble Hyperparameter Evaluation using 5-Fold Cross-Validation

Base Models	Meta Learner	Avg CV Accuracy (%)	Avg CV F1 Score
DT + KNN	LR ($C = 10$, liblinear)	92.97	0.9017
DT + KNN	LR ($C = 0.1$, lbfgs)	92.97	0.9020
DT + KNN	LR ($C = 1$, lbfgs)	92.75	0.9002
DT + KNN	LR ($C = 0.01$, liblinear)	93.19	0.9089

7. Performance Comparison

Table 4: Performance Comparison of Ensemble Models on Test Set

Model	Accuracy (%)	Precision	Recall	F1 Score
Bagging	97.37	1.00	0.93	0.963
AdaBoost	98.25	1.00	0.95	0.976
Gradient Boosting	96.49	1.00	0.90	0.950
Stacking	92.98	0.97	0.83	0.897

Analysis:

- **AdaBoost** achieved the highest accuracy (98.25%) and F1 score, making it the best-performing model.
- **Bagging** also performed very well with high stability and strong recall.
- **Gradient Boosting** showed slightly lower recall, leading to a marginal drop in F1 score.
- **Stacking** had the lowest performance due to lower recall, indicating missed positive cases.

8. Observation Questions

1. How does Bagging reduce variance?

Bagging (Bootstrap Aggregating) reduces variance by training multiple models on different bootstrap samples of the dataset. Each model learns from a slightly different subset of data, leading to diverse predictions. By aggregating these predictions (majority voting), the overall variance is reduced, resulting in a more stable and less overfitted model.

2. How does Boosting address model bias?

Boosting reduces bias by training models sequentially, where each new model focuses on correcting the errors made by previous models. Misclassified samples are given higher importance, allowing the model to learn difficult patterns. This iterative improvement helps convert weak learners into a strong learner.

3. Why does stacking benefit from heterogeneous models?

Stacking combines different types of base models (heterogeneous models) that capture different patterns in the data. Since each model has its own strengths and weaknesses, combining them improves overall performance. A meta-learner learns how to optimally combine their predictions, enhancing generalization.

4. Which ensemble method performed best and why?

In this experiment, Bagging and AdaBoost performed the best with the highest accuracy of approximately 97.37%. Bagging effectively reduced variance by averaging multiple models, while AdaBoost reduced bias by focusing on misclassified samples. Gradient Boosting also performed well but slightly lower. Stacking showed comparatively lower performance due to limited diversity among base models.

9. Conclusion

In this experiment, ensemble learning techniques including Bagging, AdaBoost, Gradient Boosting, and Stacked Ensemble were implemented on the Wisconsin Diagnostic Breast Cancer dataset and evaluated using both cross-validation and test data.

Among all models, **AdaBoost achieved the best performance** with an accuracy of **98.25%** and the highest F1 score (0.976), demonstrating its effectiveness in reducing bias by focusing on misclassified samples. **Bagging** also performed strongly with an accuracy of **97.37%**, showing its ability to reduce variance and improve model stability.

Gradient Boosting achieved competitive performance (**96.49% accuracy**), but slightly lower recall led to a minor drop in F1 score. The **Stacked Ensemble** model showed comparatively lower performance (**92.98% accuracy**), likely due to limited diversity among base models and less effective meta-learning.

Overall, the experiment demonstrates that ensemble methods significantly improve predictive performance compared to individual models. Bagging is effective for variance reduction, Boosting is powerful for bias reduction, and Stacking can be effective when diverse models are properly combined and tuned.